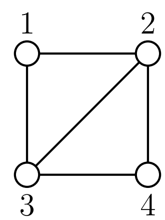
Indian Statistical Institute

BSDS - Second Year

Exercise Set 5.

1. Consider a random walk on the following graph:



   a) Write the transition matrix.

   b) If the walker starts at vertex 1, i.e., $\mathbb{P}(X_0 = 1) = 1$, what is the probability distribution after 2 steps, i.e., find $\mathbb{P}(X_2 = i)$ for $i = 1, 2, 3, 4$?

   c) Is this chain doubly stochastic? Explain.

2. For any Markov chain $\{X_n : n \geq 0\}$ with transition matrix $\mathbf{P}$, show that $\{Y_n := X_{nk} : n \geq 0\}$ is also a Markov chain for any $k \geq 2$. Find its transition matrix. Show by an example that the chain $\{Y_n : n \geq 0\}$ may not irreducible even if the original chain is irreducible.

3. Consider a Markov chain on the state space $\mathcal{S} = \{1, 2, 3, 4\}$ with transition matrix

$$\mathbf{P} = \begin{array}{c c} & \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{3}{4} & 0 & 0 & \frac{1}{4} \\ \frac{2}{3} & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{2} & \frac{2}{3} & 0 \end{bmatrix} \end{array}.$$

   a) Find a recurrence relation between $p_{11}^{(n)}$ and hence find $p_{11}^{(n)}$.

   b) Compute $f_{11}^{(n)}$. Show that the state is positive recurrent. Compute $\mu_1 = \sum_{n=0}^{\infty} n f_{11}^{(n)}$.

4. Consider the random walk on a graph $G = (V, E)$. Let $i, j \in V$.

a) For $i_1, i_2, \ldots, i_{n-1} \in V$, show that

$$\begin{aligned}&\deg(i)\mathbb{P}(X_1 = i_1, X_2 = i_2, \ldots, X_{n-1} = i_{n-1}, X_n = j \mid X_0 = i)\\&= \deg(j)\mathbb{P}(X_1 = i_{n-1}, X_{n-2} = i_2, \ldots, X_{n-1} = i_1, X_n = i \mid X_0 = j)\end{aligned}$$

where $\deg(u)$ is the degree of the vertex u.

b) For any $n \geq 1$, show that

$$\deg(i)p_{ij}^{(n)} = \deg(j)p_{ji}^{(n)}.$$

5. For any $i, j \in \mathcal{S}$, show that

$$\sup_{n\geq 1} p_{ij}^{(n)} \leq f_{ij}^{\star} \leq \sum_{n=1}^{\infty} p_{ij}^{(n)}.$$

6. Suppose that $i \rightsquigarrow j$. Define the event E_{ij} that chain visits j before returning to i starting from i. More precisely, for $n \geq 1$, set

$$g_{ij}^{(n)} = \mathbb{P}(X_n = j, X_t \neq i \text{ for } 1 \leq t < n \mid X_0 = i).$$

Show that $\mathbb{P}(E_{ij} \mid X_0 = i) = \sum_{n=1}^{\infty} g_{ij}^{(n)} > 0$.

7. Suppose $\{\epsilon_i : i \geq 1\}$ be **iid** random variable taking values in $\mathbb{Z}$ such that $\mathbb{E}(\epsilon_1) \neq 0$. Let $S_0 = 0$ and $S_n = \sum_{i=1}^{n} \epsilon_i$ for $n \geq 1$. Show that the chain $\{S_n : n \geq 0\}$ must be transient.

[Hint: Strong law of large numbers.]

8. For states i, j and any $n, k_0 \geq 1$, show that

$$\begin{aligned}&\mathbb{P}(X_t \neq j \text{ for } 1 \leq t \leq n + k_0 \mid X_0 = i)\\&\leq \mathbb{P}(X_t \neq j \text{ for } 1 \leq t \leq n \mid X_0 = i) \sup\big\{\mathbb{P}(X_t \neq j \text{ for } 1 \leq t \leq k_0 \mid X_0 = k) : k \neq j\big\}.\end{aligned}$$

9. For states $i, j \in \mathcal{S}$ with $i \rightsquigarrow j$ in a Markov chain having N states show that there exists $n \leq N - 1$ such that $p_{ij}^{(n)} > 0$.

10. For a positive recurrent state i, show that

$$\mathbb{E}(\tau_i \mid X_0 = j) < \infty$$

for any j such that $i \rightsquigarrow j$.

11. Consider an irreducible aperiodic Markov chain.

a) For any pair of states $i, j \in \mathcal{S}$, show that there exists $M = M(i, j)$ such that $p_{ij}^{(n)} > 0$ for all $n \geq M$.

b) If the chain has N states, show that there exists $M = M(N)$ such that $p_{ij}^{(n)} > 0$ for all states i, j and $n \geq M$.

c) For Exercise 1, explicitly come up with a value M such that $p_{ij}^{(n)} > 0$ for any $i, j \in \mathcal{S}$ and $n \geq M$.

12. Suppose $\{X_n : n \geq 0\}$ is a Markov chain. For some integer $K \geq 1$ and a state i_0, suppose that

$$\inf\{p_{i,i_0}^{(K)} : i \in \mathcal{S}\} > 0.$$

Show that,

$$\mathbb{P}(\tau_{i_0} > n \mid X_0 = i) \leq C_1 \exp(-C_2 n)$$

for some $C_1, C_2 > 0$ and for any $i \in \mathcal{S}$ where

$$\tau_k = \inf\{n \geq 1 : X_n = k\}.$$